

# Statistics: Preliminary Ph.D. Course

## Day 2: Random Variables

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# Today's Session

We will more or less follow the structure in Hansen's Book

- Univariate **Discrete Random Variables**. Definition and Examples. Support and PMFs. Expectation Operator and Linearity. Definitions, Theorems, Proofs, and Examples.
- Univariate **Continuous Random Variables**. Support, PDFs, and Transformations. Expectation Operator and Linearity. Definitions, Theorems, Proofs, and Examples.
- The **Distribution Function** (CDF): Definition and Properties. Relation to Axioms of Probability.
- **Moments** and the Variance Operator: Definition, Properties, and Examples.
- **Jensen's Inequality**: Theorem, Proof, and Examples.
- **Symmetric Distributions**: Definition, Propositions, Proofs, and Examples.
- **MGF** and **ChF**: Definition, Theorems, Proofs, and Examples.
- **Generalization** to the Bivariate and Multivariate cases. Covariances.

# Random Variables

## Definition

- It is convenient to represent random outcomes numerically. *Example.*
- **Definition 2.1:** A (univariate) **Random Variable**  $X$  is a real-valued outcome, i.e., a function from the sample space  $\Omega$  to the real line  $\mathbb{R}$ :

$$X(\omega) : \Omega \longrightarrow \mathbb{R}$$

$\omega \mapsto X$

- Some random variables take values only on subsets of the real line. *Examples.*
- **Definition 2.2:** A set  $\mathcal{X}$  is **Discrete** if it has a finite or countably infinite number of elements. *Examples.*
- **Definition 2.3:** If there is a discrete set  $\mathcal{X}$  such that  $\mathbb{P}(X \in \mathcal{X}) = 1$ , then  $X$  is a **Discrete Random Variable**.

# Discrete Case: Support, PMF, and Expectation

## Definitions

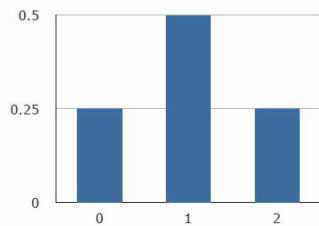
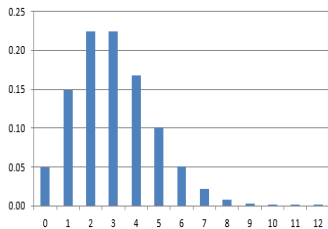
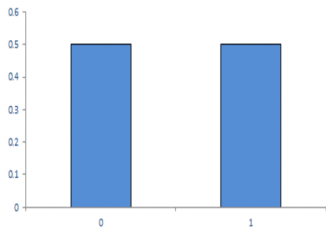
- **Definition 2.4:** The smallest set  $\mathcal{X}$  with the property  $\mathbb{P}(X \in \mathcal{X}) = 1$  is the **Support** of the random variable  $X$ . It is the set of values with a strictly positive probability of occurrence:  $\mathcal{X} = \{\tau_1, \tau_2, \dots\}$ . We call the points  $\tau_j$  the **Support Points**.
- **Definition 2.5:** The **Probability Mass Function (PMF)** of a discrete random variable  $X$  is  $\pi(x) = \mathbb{P}(X = x)$ , the probability that  $X$  takes the value  $x$ . When evaluated at the support points  $\tau_j$  we write  $\pi_j = \pi(\tau_j)$ .
- A necessary condition for a valid PMF is  $\sum_j \pi_j = 1$ . *Examples.*
- **Definition 2.6:** For a discrete random variable  $X$  with support points  $\tau_j$ , the **Expectation** of  $X$  is:

$$\mathbb{E}[X] = \sum_j \tau_j \pi(\tau_j)$$

*Examples.*

# Probability Mass Functions

## An Illustration



# Discrete Case: The Expectations Operator

## Properties

- **Definition 2.7:** For a discrete random variable  $X$  with support points  $\tau_j$ , the **Expectation** of the transformation  $g(X)$  is:

$$\mathbb{E}[g(X)] = \sum_j g(\tau_j)\pi(\tau_j)$$

## Theorem 2.1 (Linearity of Expectations)

For any constants  $a, b \in \mathbb{R}$ ,  $\mathbb{E}[a + bX] = a + b\mathbb{E}[X]$ .

**Proof.** (Hint: Use the definition of expectation and the basic properties of sums).

- Expectations can be finite, infinite, or undefined, depending on the series  $\sum_j$ . If it is not convergent, it may be either infinite or undefined.

*Examples.*

# The Distribution Function

- Any random variable (either continuous or discrete)  $X$  can be represented by its distribution function:  $X \sim F$ .
- **Definition 2.8:** The **Distribution Function** is  $F(x) = \mathbb{P}(X \leq x)$ , i.e., the probability of the event  $\{X \leq x\}$ . It is also called the **Cumulative Distribution Function (CDF)**.
- For the discrete case, the CDF is:  $F(\tau_k) = \mathbb{P}(X \leq \tau_k) = \sum_{j=1}^k \pi(\tau_j)$ .

## Theorem 2.2 (Properties of a CDF)

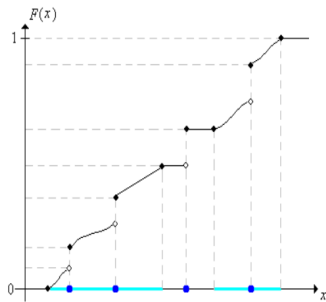
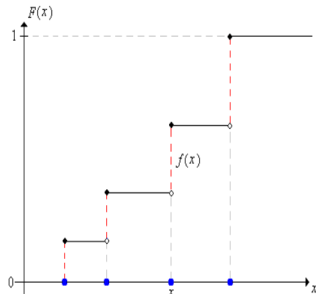
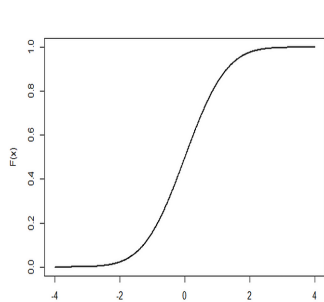
Let  $F(x)$  be a distribution function,  $F : \mathbb{R} \rightarrow [0, 1]$ . Then:

1.  $F(x)$  is non-decreasing.
2.  $\lim_{x \rightarrow -\infty} F(x) = 0$ .
3.  $\lim_{x \rightarrow +\infty} F(x) = 1$ .
4.  $F(x)$  is right-continuous:  $\lim_{x \rightarrow x_0^+} F(x) = F(x_0)$ .

*Relation to Axioms. Cadlag Functions.*

# The Distribution Function

## An Illustration





# Continuous Case: PDF, Support, and Expectation

## Definitions

- If a random variable  $X$  takes a continuum of values it is not discretely distributed.
- **Definition 2.9:** If  $X \sim F$ , and  $F(x)$  is continuous, then  $X$  is a **Continuous Random Variable**.
- An **important property** is that the probability that  $X$  equals any specific value is zero:  $\mathbb{P}(X = x) = 0$ .  
*Proof.* (Hint: Take the event  $[x, x + \epsilon]$ , use the definition of CDF, and take the limit as  $\epsilon \rightarrow 0$ ).

## Examples.

- Continuous random variables do not have PMF. An analog is the derivative<sup>1</sup> of  $F$ .
- **Definition 2.10:** When  $F(x)$  is differentiable, its **Probability Density Function (PDF)** is

$$f(x) = \frac{d}{dx}F(x)$$

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<sup>1</sup>More on this later, in the Measure Theory part.

# Continuous Case: PDF, Support, and Expectation

## Definitions

### Theorem 2.3 (Properties of a PDF)

A function  $f(x)$  is a density function if and only if:

1.  $f(x) \geq 0 \ \forall x$ .
2.  $\int_{\mathbb{R}} f(x) \, dx = 1$ .

*Examples.*

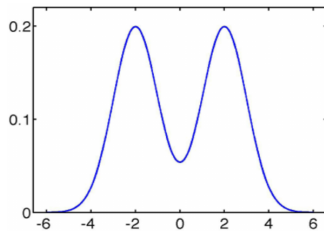
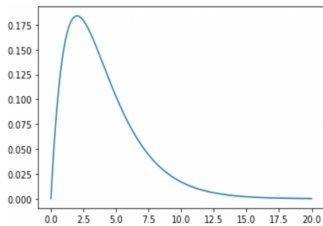
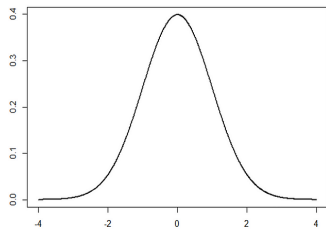
- We can use densities to compute probabilities of events (sets):

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x) \, dx$$

- **Definition 2.11:** The **Support**  $\mathcal{X}$  of a continuous random variable is the smallest set containing  $\{x : f(x) > 0\}$ .

# Probability Density Functions

## An Illustration



# Continuous Case: PDF, Support, and Expectation

## Definitions

- **Definition 2.12:** If  $X$  is continuously distributed with density  $f(x)$  its **Expectation** is defined as:

$$\mathbb{E}[X] = \int_{\mathbb{R}} x f(x) dx$$

- **Definition 2.13:** If  $X$  has density  $f(x)$  then the **Expectation** of  $g(X)$  is:

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) f(x) dx$$

*Examples. Finiteness vs Infiniteness vs Non-existence.*

## Theorem 2.4 (Linearity of Expectations)

For any constants  $a, b \in \mathbb{R}$ ,  $\mathbb{E}[a + bX] = a + b\mathbb{E}[X]$ .

**Proof.** (Hint: Use the definition of expectation and the properties of integrals).

# Transformations

## Theorem and Applications

A natural question is: If  $X \sim F$  for some known  $F$ , what is the distribution of  $Y = g(X)$ ?

### Theorem 2.5 (Distributions of Monotonic Transformations)

If  $X \sim f_X(x)$ ,  $f(x)$  is continuous on  $\mathcal{X}$ ,  $g(x)$  is strictly monotone, and  $g^{-1}(y)$  is continuously differentiable on  $\mathcal{Y}$ , then for  $y \in \mathcal{Y}$ :

$$f_Y(y) = f_X(g^{-1}(y)) |J(y)|$$

where  $J(y) = \frac{d}{dy}g^{-1}(y)$ .

*Applications.*

*Probability Integral Transformations (PIT).*

*Examples of Non-Monotonic Transformations.*

# The Variance Operator

## Definitions, Properties, and Examples

- **Definition 2.14:** The **Variance** of  $X$  is  $\sigma^2 = \text{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2]$ .
- The variance is necessarily greater than zero, and either finite or infinite. It is zero only for degenerate random variables.
- **Definition 2.15:** A random variable is **Degenerate** if for some value  $c \in \mathbb{R}$ ,  $\mathbb{P}(X = c) = 1$ . It is essentially non-random.
- **Definition 2.16:** The **Standard Deviation** of  $X$  is the positive square root of the variance:  $\sigma = \sqrt{\sigma^2}$ .
- It is typical to use the mean and standard deviation to summarize the center and spread of a distribution.

# The Variance Operator

## Definitions, Properties, and Examples

### Theorem 2.6

$$\mathbb{V}ar[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

***Proof.***

### Theorem 2.7 (Quadratic Property of the Variance Operator)

For any constants  $a, b \in \mathbb{R}$ :

$$\mathbb{V}ar[a + bX] = b^2 \mathbb{V}ar[X]$$

***Proof.*** (Hint: Use the definition of variance).

*Examples.*

# Moments of Random Variables

## Definition and Examples

- **Definition 2.17:** The  $m$ -th moment of a random variable  $X$  is  $\mathbb{E}[X^m]$ .
- For non-negative  $X$ ,  $m$  can be real-valued. Even moments are either finite or undefined. Odd moments may be finite, infinite, or undefined.
- An important result is that if  $\mathbb{E}[|X|^r] < \infty$  then  $\mathbb{E}[|X|^s] < \infty \forall 0 \leq s \leq r$ . (See Lyapunov's Inequality in Hansen's Book pg. 43).

*Examples.*



# Jensen's Inequality

## Theorem, Proof, and Examples

### Theorem 2.8 (Jensen's Inequality)

For any random variable  $X$ , if  $g(\cdot)$  is a convex function then:

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)]$$

If  $g(\cdot)$  is a concave function, then:

$$g(\mathbb{E}[X]) \geq \mathbb{E}[g(X)]$$

**Proof (Convex Case).** (Hint: Use the tangent line to  $g(x)$  at  $x = \mathbb{E}[X]$  and take expectations).

*Examples.*

*Applications.*

# Symmetric Distributions

## Definition

- **Definition 2.18:** A random variable has a **Symmetric Distribution** about 0 if the distribution function  $F$  satisfies:

$$F(x) = 1 - F(-x)$$

Furthermore, if  $X$  has a density  $f(x)$ :

$$f(x) = f(-x)$$

i.e., if its density is an odd function.

- There are some interesting results for symmetric distributions.

# Symmetric Distributions

## Results

### Proposition 2.1 (Odd Moments of Symmetric Distributions)

If a distribution is symmetric about zero then all odd moments are zero if the moment is finite.

**Proof.** (Hint: Assume that the density exists, use the definition of moment, integrate separately on the positive and negative reals, and use the definition of symmetry).

### Theorem 2.9 (Odd Functions and Symmetric Distributions)

If  $f(x)$  is symmetric about zero,  $g(x)$  is odd, and  $\int_{\mathbb{R}_+} |g(x)|f(x)dx < \infty$ , then  $\mathbb{E}[g(X)] = 0$

**Proof.** (Hint: Use the definition of expectation, integrate separately on the positive and negative reals, and use the definition of odd function).

# The Moment Generating Function

## Definition and Theorems

- **Definition 2.19:** The **Moment Generating Function (MGF)** of  $X$  is  $M(t) = \mathbb{E}[\exp(tX)]$ .

*Examples.*

### Theorem 2.10 (Moments and the MGF)

If  $M(t)$  is finite for  $t$  in a neighborhood of 0 then  $M(0) = 1$  and:

$$\left. \frac{d^m}{dt^m} M(t) \right|_{t=0} = \mathbb{E}[X^m]$$

**Proof (Continuous Case).** (Hint: Use the definition of expectation and assume you can differentiate under the integral sign).

*Examples.*

# The Characteristic Function (ChF)

## Definition and Theorems

- Since the MGF may not be finite, the ChF may be used instead for advanced proofs.
- **Definition 2.20:** The **Characteristic Function (ChF)** of  $X$  is  $Ch(t) = \mathbb{E}[\exp(itX)]$ , where  $i = \sqrt{-1}$ .

## Theorem 2.11 (Moments and the ChF)

The characteristic function satisfies:  $Ch(0) = 1$  and  $\left. \frac{d^m}{dt^m} Ch(t) \right|_{t=0} = i^m \mathbb{E}[X^m]$ .

**Proof (Continuous Case).** (Hint: The same as for the MGF).

## Theorem 2.12 (Symmetry and the ChF)

If  $X$  is symmetrically distributed about zero its characteristic function equals  $Ch(t) = \mathbb{E}[\cos(tX)]$ .

**Proof.** (Hint: Use Euler's Formula  $\exp(iu) = \cos(u) + i \sin(u)$  and Theorem 2.9).

# Bivariate Random Vectors

## Definitions

- We can **generalize** the previous discussion to higher dimensions,  $\mathbb{R}^m$ . We will focus on the bivariate case  $m = 2$ .
- **Definition 2.21:** A bivariate **Random Vector**  $(X, Y)'$  is a pair of real-valued outcomes, i.e., a function from the sample space  $\Omega$  to  $\mathbb{R}^2$ :

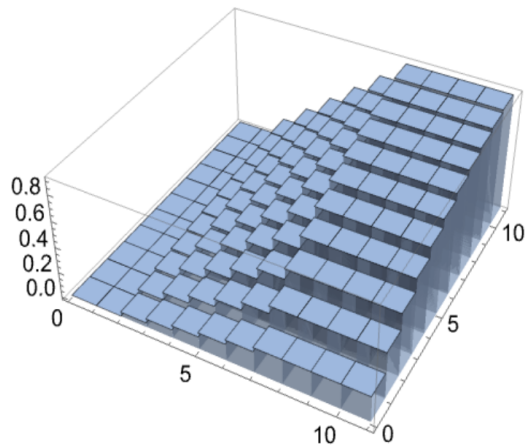
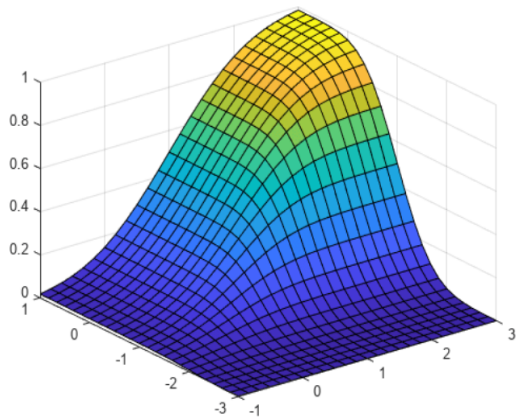
$$\begin{pmatrix} X \\ Y \end{pmatrix} : \Omega_1 \times \Omega_2 \longrightarrow \mathbb{R}^2$$
$$(\omega_1, \omega_2) \mapsto (x, y)$$

- **Definition 2.22:** The **Joint Distribution Function** of the bivariate random vector  $(X, Y)'$  is  $F(x, y) = \mathbb{P}(X \leq x, Y \leq y) = \mathbb{P}(\{X \leq x\} \cap \{Y \leq y\})$ . (Same properties).

*Examples.*

# Bivariate Random Vectors

## Joint Distribution Function: An Illustration



# Bivariate Random Vectors

## Joint PMFs

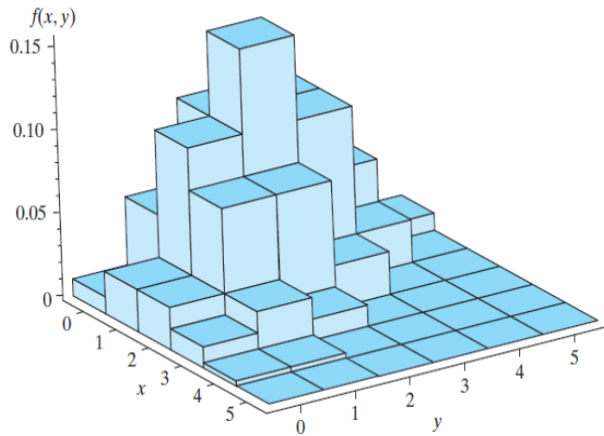
- **Definition 2.23:** A pair of random variables is **Discrete** if there is a discrete set  $\mathcal{S} \subseteq \mathbb{R}^2$  such that  $\mathbb{P}((X, Y)' \in \mathcal{S})$ .
- **Definition 2.24:** The set  $\mathcal{S} \subseteq \mathbb{R}^2$  is the **Support** of  $(X, Y)$  and consists of a set of points in  $\mathbb{R}^2$ . In many cases  $\mathcal{S} = \mathcal{X} \times \mathcal{Y}$ , where  $\mathcal{X} \subseteq \mathbb{R}$  and  $\mathcal{Y} \subseteq \mathbb{R}$ . And we can write the **Support Points** as  $\{\tau_1^x, \tau_2^x, \dots\}$  and  $\{\tau_1^y, \tau_2^y, \dots\}$ .
- **Definition 2.25:** The **Joint Probability Mass Function (PMF)** of the bivariate random vector  $(X, Y)'$  is  $\pi(x, y) = \mathbb{P}(X = x, Y = y)$ . At the support points we write  $\pi_{ij} = \pi(\tau_i^x, \tau_j^y)$ .
- A necessary condition for a valid Joint PMF is  $\sum_i \sum_j \pi_{ij} = 1$ .

*Example.*



# Bivariate Random Vectors

## Joint PMF: An Illustration



# Bivariate Random Vectors

## Joint PDFs

- The pair  $(X, Y)'$  is continuously distributed if its CDF  $F(x, y)$  is continuous in  $(x, y)$ .
- **Definition 2.26:** When  $F(x, y)$  is differentiable, its **Joint Probability Density Function (PDF)**,  $f(x, y)$  is

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

They are non-negative functions that integrate to one in  $\mathbb{R}^2$  (these are the two conditions for a valid joint density).

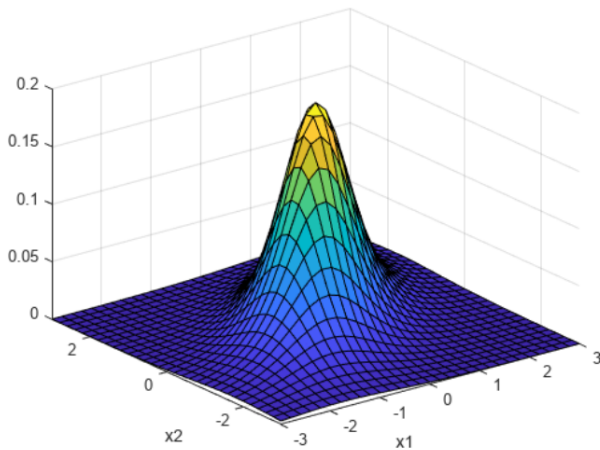
- As in the univariate case, we can use them to compute probabilities:

$$\mathbb{P}(a \leq X \leq b, c \leq Y \leq d) = \int_c^d \int_a^b f(x, y) \, dx dy$$

*Examples.*

# Bivariate Random Vectors

## Joint PDF: An Illustration



# Bivariate Random Vectors

## Marginal Distributions

- The joint distribution of  $(X, Y)'$  fully describes the distribution of each component of the random vector.
- **Definition 2.27:** Let  $(X, Y)'$  be a bivariate random vector. The Marginal Distribution of  $X$  is  $F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}(X \leq x, Y \leq \infty) = \lim_{y \rightarrow +\infty} F(x, y)$ .
- In the continuous and discrete cases we write, respectively:

$$F_X(x) = \lim_{y \rightarrow +\infty} \int_{-\infty}^y \int_{-\infty}^x f(x, y) \, dx dy = \int_{\mathbb{R}} \int_{-\infty}^x f(x, y) \, dx dy$$

$$F_X(x) = \lim_{\tau_k \rightarrow +\infty} \sum_{j=1}^{\tau_k} \sum_{i=1}^{\tau_m} \pi_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\tau_m} \pi_{ij} = \sum_j \sum_{i=1}^{\tau_m} \pi_{ij}$$

Examples.

# Bivariate Random Vectors

## Marginal Densities and Expectations

- **Definition 2.28:** The **Marginal Density** of  $X$  and that of  $Y$  given a joint density  $f(x, y)$  are:

$$f_X(x) = \int_{\mathbb{R}} f(x, y) dy \quad \text{and} \quad f_Y(y) = \int_{\mathbb{R}} f(x, y) dx$$

- **Definition 2.29:** Let  $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ . Then:

$$\mathbb{E}[(X, Y)'] = (\mathbb{E}[X], \mathbb{E}[Y])'$$

$$\mathbb{E}[g(X, Y)] = \int_{\mathbb{R}} \int_{\mathbb{R}} g(x, y) f(x, y) dx dy \quad \text{in the continuous case}$$

$$\sum_{(x, y) \in \mathbb{R}^2 : \pi(x, y) > 0} g(x, y) \pi(x, y) \quad \text{in the discrete case}$$

*Examples.*

# Bivariate Random Vectors

## Independence

- Recall the definition of independence:  $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ . Consider the events  $A = \{X \leq x\}$  and  $B = \{Y \leq y\}$ . So that  $\mathbb{P}(A \cap B) = \mathbb{P}(\{X \leq x\} \cap \{Y \leq y\}) = F(x, y)$ .
- **Definition 2.30:** The random variables  $X$  and  $Y$  are said to be **Statistically Independent**, written  $X \perp\!\!\!\perp Y$  or  $X \perp Y$ , if for all  $x$  and  $y$ :  $F(x, y) = F_X(x)F_Y(y)$ .

### Theorem 2.13 (Independence with PDFs and PMFs)

The discrete random variables  $X$  and  $Y$  are independent if  $\forall x, y, \pi(x, y) = \pi(x)\pi(y)$ . Moreover, if  $X$  and  $Y$  have a continuous and differentiable joint distribution function, they are statistically independent if  $\forall x, y, f(x, y) = f_X(x)f_Y(y)$ .

**Proof.** (Hint: Use the definition of independence and differentiate).

# Bivariate Random Vectors

## Independence

### Theorem 2.14 (Independence and MGFs)

If  $X$  and  $Y$  are independent with MGFs  $M_X(t)$  and  $M_Y(t)$ , then the MGF of  $Z = X + Y$  is  $M_Z = M_X(t)M_Y(t)$ .

**Proof.** (Hint: Use the definitions of MGF and Theorem 2.13).

- In fact, we can prove the following:

### Theorem 2.15 (Independence and Expectations)

If  $X$  and  $Y$  are statistically independent then for any two functions  $g : \mathbb{R} \rightarrow \mathbb{R}$  and  $h : \mathbb{R} \rightarrow \mathbb{R}$  such that  $\mathbb{E}[|g(X)|] < \infty$  and  $\mathbb{E}[|h(Y)|] < \infty$ ,  $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$ .

**Proof.** (Hint: Use the definition of independence and Theorem 2.13).

# Bivariate Random Vectors

## Independence

### Theorem 2.16 (Independence and Transformations)

If  $X$  and  $Y$  are statistically independent then for any two functions  $g : \mathbb{R} \rightarrow \mathbb{R}$  and  $h : \mathbb{R} \rightarrow \mathbb{R}$ ,  $U = g(X)$  and  $V = h(Y)$  are independent.

**Proof.** (Hint: Use the definition of independence and define the pre-image sets  $A(u) = \{x : g(x) \leq u\}$  and  $B(v) = \{y : h(y) \leq v\}$ ).

*Examples.*



# Bivariate Random Vectors

## Covariance and Correlation

- **Definition 2.31:** The **Covariance** between  $X$  and  $Y$  is

$$\mathbb{C}ov(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

- **Definition 2.32:** The **Correlation** between  $X$  and  $Y$  is

$$\rho(X, Y) = \frac{\mathbb{C}ov(X, Y)}{\sqrt{\mathbb{V}ar[X]}\sqrt{\mathbb{V}ar[Y]}}$$

- If  $\mathbb{C}ov(X, Y) = 0$  the  $\rho(X, Y) = 0$ , and it is typical to say that  $X$  and  $Y$  are uncorrelated.

## Theorem 2.17 (Independence and Correlation)

If two random variables  $X$  and  $Y$  are statistically independent with finite variances, they are uncorrelated. The reverse is not true.

**Proof.** (Hint: Use the definitions of covariance and independence, then find a counterexample of two dependent random variables to prove that the reverse is not true. Think about symmetry).

# Bivariate Random Vectors

## Covariance and Correlation

### Theorem 2.18 (Property of the Variance)

If  $X$  and  $Y$  have finite variances then  $\mathbb{V}ar[X \pm Y] = \mathbb{V}ar[X] + \mathbb{V}ar[Y] \pm 2\mathbb{C}ov(X, Y)$ .

### Corollary 2.18

If  $X$  and  $Y$  are uncorrelated and have finite variances then  $\mathbb{V}ar[X \pm Y] = \mathbb{V}ar[X] + \mathbb{V}ar[Y]$ .

*Examples.*

# Multivariate Random Vectors

## Key Definitions

- We now consider the case of an  $m \times 1$  random vector  $X \in \mathbb{R}^m$ :  $(X_1, \dots, X_m)'$ .
- **Definition 2.33:** The **Joint Distribution Function** is  $F(x) = \mathbb{P}(X_1 \leq x_1, \dots, X_m \leq x_m)$ .
- **Definition 2.34:** The **Joint PMF** is  $\pi(x) = \mathbb{P}(X \leq x) = \mathbb{P}(X_1 = x_1, \dots, X_m = x_m)$ .
- **Definition 2.35:** When  $F(x)$  is differentiable, its **Joint PDF** equals:

$$f(x) = \frac{\partial^m}{\partial x_1 \dots \partial x_m} F(x)$$

- **Definition 2.36:** The **Expectation** of  $X$  given by the vector of expectations:

$$\mathbb{E}[X] = \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_m] \end{pmatrix}$$

# Multivariate Random Vectors

## Key Definitions

- **Definition 2.37:** The  $m \times m$  **Variance-Covariance Matrix**  $\Sigma$  of  $X$  is given by

$$\mathbb{V}ar[X] = \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])'] = \mathbb{E}[XX'] - \mathbb{E}[X]\mathbb{E}[X]' = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1m} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{m1} & \sigma_{m2} & \cdots & \sigma_m^2 \end{pmatrix}$$

where  $\sigma_j^2 = \mathbb{V}ar[X_j]$  and  $\sigma_{ij} = \mathbb{C}ov(X_i, X_j)$  for  $i, j = 1, \dots, m$  and  $i \neq j$ .

## Theorem 2.19 (Properties of the Var-Cov Matrix)

1.  $\Sigma$  is Symmetric.
2.  $\Sigma$  is Positive Semidefinite (p.s.d.).

**Proof.** (Hint: Use the definition of symmetric matrices, then show that the quadratic form  $a'\Sigma a \geq 0$  for some  $a \in \mathbb{R}^m$ ).

# Multivariate Random Vectors

## Key Definitions

### Theorem 2.20 (Var-Cov Matrix of Linear Combinations)

If the random vector  $X$  with  $m \times 1$  expectation  $\mu$  and  $m \times m$  variance-covariance matrix  $\Sigma$  and  $A$  is a  $q \times m$  matrix, then  $AX$  is a  $q \times 1$  random vector with expectation  $A\mu$  and  $q \times q$  variance-covariance matrix  $A\Sigma A'$ .

**Proof.** (Hint: Use the definitions of expectations and variance-covariance matrix).

### Theorem 2.21 (Finiteness of Multivariate Expectations)

For  $X \in \mathbb{R}^m$ ,  $\mathbb{E}[\|X\|] < \infty$  if and only if  $\mathbb{E}[|X_j|] < \infty$  for  $j = 1, \dots, m$ .

**Proof.** (For the implication use the triangle inequality for the vector Euclidean norm. For the 'only if' part of the statement use the fact that the Euclidean norm of a vector is larger than the length of any individual component).

# Multivariate Random Vectors

- Finally, let's see a very useful multivariate generalization of Theorem 2.5.

## Theorem 2.22 (Distributions of Monotonic Transformations)

Let  $X \in \mathbb{R}^m$  have density  $f_X(x)$ , let  $g$  be a one-to-one function  $g : \mathbb{R}^m \rightarrow \mathbb{R}^q$ , and let  $h(y) = g^{-1}(y)$  be differentiable. Then, the random variable  $Y = g(X)$  has density:

$$f_Y(y) = f_X(h(y))|\det(J(y))|$$

where  $J(y) = \partial h(y)/\partial y'$  is the Jacobian of  $h(y)$ .

*Examples.*

# Recommended Exercises

Hansen's Book

- Chapter 2, Exercises: 2.1-2.16.
- Chapter 4, Exercises: 4.1a)-c), 4.1 e)-f), 4.2 a)-d), 4.3 a)-c), 4.4-4.12, 4.14, 4.20, 4.22.

# References

Hansen, B. (2022). Probability and Statistics for Economists. *Princeton University Press*. Chapters 2 and 4.

Wasserman, L. (2004). All of Statistics: A Concise Course in Statistical Inference. *Springer*.

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